

The ratio test for future GNSS ambiguity resolution

Sandra Verhagen · Peter J. G. Teunissen

Received: 14 June 2012 / Accepted: 29 October 2012 / Published online: 16 November 2012
 © Springer-Verlag Berlin Heidelberg 2012

Abstract The performance of the popular ambiguity ratio test is analyzed. Based on experimental and simulated data, it is demonstrated that the current usage of the ratio test with fixed critical value is not sustainable in light of the enhanced variability that future global navigation satellite system (GNSS) ambiguity resolution will bring. As its replacement, the model-driven ratio test with fixed failure rate is proposed. The characteristics of this fixed-failure rate ratio test are described, and a performance analysis is given. The relation between its critical value and various GNSS model parameters is also studied. Finally, a procedure is presented for the creation of fixed failure rate look-up tables for the critical values of the ratio test.

Keywords Integer ambiguity resolution · Ratio test · Fixed failure rate · Integer aperture estimation · GNSS

Introduction

Any carrier phase-based GNSS model with integer ambiguities can be formulated in linear(ized) form as

$$E(\mathbf{y}) = \mathbf{A}\mathbf{a} + \mathbf{B}\mathbf{b}; \quad D(\mathbf{y}) = \mathbf{Q}_{yy} \quad (1)$$

with $E(\cdot)$ and $D(\cdot)$ the expectation and dispersion operators, respectively, and where $\mathbf{y}$ is the carrier phase and code observations, $\mathbf{a}$ denotes the integer carrier phase ambiguities,

$\mathbf{b}$ represents real-valued parameters such as baseline/range and/or atmosphere, $\mathbf{A}$ and $\mathbf{B}$ are design matrices that link observations to the unknown parameters, and $\mathbf{Q}_{yy}$ is the variance–covariance matrix of phase and code observations.

The procedure for obtaining the least-squares (LS) solution of model (1) can be divided into three steps. In the first step, one discards the integer nature of the ambiguities and performs a standard least-squares adjustment. As a result, one obtains the so-called float solution of all the parameters, that is, ambiguities, baseline components and possibly additional parameters such as atmospheric delays, together with their variance–covariance matrix

$$\begin{bmatrix} \hat{\mathbf{a}} \\ \hat{\mathbf{b}} \end{bmatrix}, \begin{bmatrix} \mathbf{Q}_{\hat{\mathbf{a}}\hat{\mathbf{a}}} & \mathbf{Q}_{\hat{\mathbf{a}}\hat{\mathbf{b}}} \\ \mathbf{Q}_{\hat{\mathbf{b}}\hat{\mathbf{a}}} & \mathbf{Q}_{\hat{\mathbf{b}}\hat{\mathbf{b}}} \end{bmatrix}.$$

In the second step, the real-valued float solution of the ambiguities is further adjusted, so as to take the integer constraints into account. As a result, one obtains an integer solution for the ambiguities $\tilde{\mathbf{a}} = I(\hat{\mathbf{a}})$ with $I: \mathbb{R}^n \rightarrow \mathbb{Z}^n$.

Integer rounding, integer bootstrapping and integer least-squares are different methods for obtaining the integer solution. Integer least-square (ILS) is optimal, as it can be shown to maximize the probability of correct integer estimation (Teunissen 1999). In contrast to rounding and bootstrapping, an integer search is needed to compute the ILS solution. This can be efficiently done with the LAMBDA method (Teunissen 1995). We restrict attention to the ILS principle.

The third step consists of correcting the float solution of all other parameters by virtue of their correlation with the ambiguities. As a result, one obtains the so-called fixed solution $\tilde{\mathbf{b}} = \hat{\mathbf{b}} - \mathbf{Q}_{\hat{\mathbf{b}}\hat{\mathbf{a}}} \mathbf{Q}_{\hat{\mathbf{a}}\hat{\mathbf{a}}}^{-1} (\hat{\mathbf{a}} - \tilde{\mathbf{a}})$. The fixed solution will have an accuracy that is in accordance with the high precision of the phase data, provided ambiguity resolution has been successful.

S. Verhagen (✉) · P. J. G. Teunissen
 Delft University of Technology, Delft, The Netherlands
 e-mail: A.A.Verhagen@TUDelft.nl

P. J. G. Teunissen
 Curtin University, Perth, Australia
 e-mail: p.teunissen@curtin.edu.au

The essence of ambiguity resolution is thus to be able to realize this much higher precision of the fixed baseline estimator. Since an incorrect integer ambiguity solution may cause severe biases in the position solution or in any other of the real-valued parameters, it is important to assess the probability of correct ILS estimation. This ILS success rate, denoted as $\mathbf{P}_{s,ILS}$, depends on the model at hand and can be computed without the need for actual observations. To have confidence in the ambiguity-resolved position solution, the ILS success rate should be very close to 1, or equivalently, the ILS failure rate, $\mathbf{P}_{f,ILS} = 1 - \mathbf{P}_{s,ILS}$, should be very close to 0. However, the user cannot control the ILS failure rate for a given measurement setup, time and location. Fortunately, the popular ambiguity ratio test provides a helping hand, since it makes a reduction in the failure rate possible in principle.

The ratio test and its performance are analyzed. First, the essence of the ratio test will be explained, together with its probabilities of false alarm, failure, successful fixing and successful detection. Next, the current usage of the ratio test with fixed critical value will be discussed, followed by a performance analysis based on experimental and simulation results.

As it will be argued and demonstrated, the current usage of the ratio test is not sustainable, certainly not in light of the enhanced variability that future GNSS ambiguity resolution will bring. As its replacement, we propose the model-driven ratio test with fixed failure rate. The characteristics of this fixed-failure rate ratio test (FF-RT) will be described, followed by a performance analysis using the same data as before. We also investigate the relation between its critical value and various GNSS model parameters. This provides qualitative and quantitative insight into the behavior of the critical value for different model strengths. Finally, we describe the procedure for creating look-up tables for the critical value as function of the fixed failure rate and certain GNSS model parameters.

Ambiguity resolution with the ratio test

The current usage and performance of the ratio test are analyzed. This is done by placing the ratio test in the framework of integer aperture estimation, thereby describing its performance in terms of the probabilities of failure, false alarm, acceptance and rejection. This is followed by a performance analysis based on experimental and simulation results.

The essence of the ratio test

Ambiguity resolution with the ratio test implies that an additional step is added to the ambiguity resolution

procedure, immediately after the second step. In this additional step, the estimated integer ambiguities are used as input to decide whether or not to accept the integer solution.

Several such tests have been proposed in the literature and are currently in use (Abidin 1993; Chen 1997; Euler and Schaffrin 1990; Frei and Beutler 1990; Han and Rizos 1996a, b; Ha 1997; Landau and Euler 1992; Tiberius and De Jonge 1995; Wang et al. 1998). Examples are the ratio test, the distance test and the projector test. The theory of integer aperture (IA) estimation as introduced by Teunissen (2003, 2004, 2005) provides a unifying principle for all these tests. Each of these tests can be shown to be a member of the class of IA estimators. A review and evaluation of these tests can be found in Verhagen (2004, 2005b) and in Verhagen and Teunissen (2006). Here, the object of study is the popular ratio test.

The ratio test was introduced with the objective to decide whether the integer ambiguity solution can be considered sufficiently more likely than any other integer candidate (Euler and Schaffrin 1990). The ratio test is defined as follows. The ILS solution $\tilde{a}$ is by definition the integer minimizer of $q(a) = (\hat{a} - a)^T Q_{\hat{a}\hat{a}}^{-1} (\hat{a} - a)$. Now, let $\tilde{a}'$ be the integer vector that returns the second smallest value of the quadratic form $q(a)$. Then, the ratio test reads as

$$\text{Accept } \tilde{a} \text{ if: } \frac{q(\tilde{a}')}{q(\tilde{a})} > c, \quad (2)$$

where c is a tolerance value, to be selected by the user. Thus, only if $q(\tilde{a}')$ is sufficiently larger than $q(\tilde{a})$, the decision will be made to accept the ILS solution. Otherwise, the ILS solution is rejected in favor of the float solution.

The ratio test in fact tests the closeness of the float solution to its nearest integer vector. If it is close enough, the test leads to acceptance of $\tilde{a}$. If it is not close enough, then the test leads to rejection in favor of the float solution $\hat{a}$. As is shown in Teunissen and Verhagen (2009), the inequality (2) describes the z -translational invariant aperture pull-in regions of the ratio test (the red and dark-green areas in bottom panel of Fig. 1). The integer solution is accepted if the float solution is inside the aperture pull-in region, otherwise it will be rejected. These aperture pull-in regions have the same size and shape and are centered at integer grid points $z \in \mathbb{Z}^n$.

The size or aperture of the pull-in region determines the smallest ratio $q(\tilde{a}')/q(\tilde{a})$ one is willing to accept. The value for c can be used to tune this aperture. In case $c = 1$, the aperture pull-in regions become equal to the ILS pull-in regions, and thus the integer solution is always accepted. In

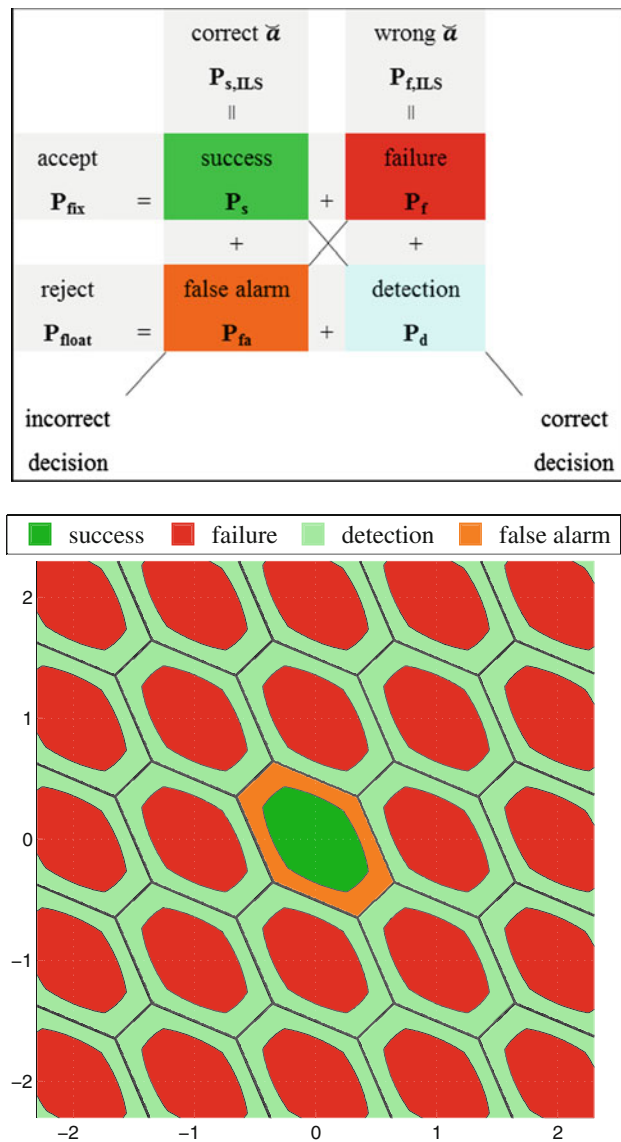


Fig. 1 Definition of *success*, *failure*, *false alarm* and *detection* probabilities, and their relation with the *fix* and *float* probabilities and the *ILS* success and failure rates. *Top panel* shows the decision matrix and the associated probabilities. *Bottom panel* shows two-dimensional example of ILS pull-in regions and acceptance regions. The colors refer to the four decision types

such case, the ratio test would be obsolete and can be discarded.

Figure 1, bottom panel, shows a two-dimensional example of the aperture pull-in regions together with the ILS pull-in regions. The aperture pull-in regions are colored red and dark green, in order to discriminate between success and failure: the correct integer in this case is the zero-vector; if the float solution resides in the dark-green aperture pull-in region, it will be correctly fixed, that is, a *success*; if it resides in one of the red regions, it will be wrongly accepted, that is, a *failure*. However, there are two

more possibilities. First, the fixed solution may be unnecessarily rejected, that is, a *false alarm*. This will be the case if the float solution resides outside the aperture pull-in regions, but inside the ILS pull-in region. This region is colored orange. Second, there is the possibility of *detection*: if the ILS solution is correctly rejected. The corresponding region is colored light green.

The top panel of Fig. 1 summarizes the four types of decisions and distinguishes between the correct and incorrect decisions. The corresponding probabilities and their relations are depicted as well. The probability that the ratio test accepts the ILS solution is given by the fix probability P_{fix} . This fix probability is the sum of the probability of correct fixing P_s and the probability of incorrect fixing P_f , $P_{fix} = P_s + P_f$.

The complement to the fix probability is the float probability P_{float} . It is equal to the sum of the false alarm probability P_{fa} and the detection probability P_d , $P_{float} = P_{fa} + P_d$.

From Fig. 1, we can now deduce the following relations:

$$c \blacktriangle \Rightarrow P_{fix} \blacktriangledown, P_f \blacktriangledown, \text{ and } P_{fa} \blacktriangle \quad (3)$$

where $\blacktriangle$ denotes an increase and $\blacktriangledown$ a decrease. Hence, for the same probability density function of the ambiguity float solution, an increase in the c -value (smaller aperture and therefore smaller acceptance area) implies a higher false alarm rate, a lower failure rate and a lower fix probability.

Ratio test and biases

Although biases in the float solution may affect the outcome of the ratio test, the ratio test is not developed to test for misspecifications in the assumed functional and/or stochastic model. Hence, the ratio test is not designed to test for the presence of biases, as is sometimes incorrectly assumed. In fact, it is easily verified that if a bias would be integer-valued, the ratio test is invariant for such a bias. Therefore, standard hypothesis testing should always be applied before proceeding with ambiguity resolution to test for inconsistencies in the data and/or model, in order to prevent that certain biases propagate in the ambiguity solution.

The ratio test and its current usage

The question is now how to choose the critical value c . Different values have been proposed in the literature, all based on empirical results, that is, those values seemed to give good performance for the given data and measurements considered. The values reported are 1.5, 2 and 3 by Han and Rizos (1996a, b), Wei and Schwarz (1995) and

Leick (2003), respectively. Recently, Parkins (2011) used the ratio test with a value of 2.5 and concluded that this value might be appropriate for shorter baselines, but that a higher value may need to be used for longer baselines. The different values are already an indication that there is not one specific value that will always give the best performance, as will be illustrated with the following examples.

Example 1: short kinematic baseline

This example is taken from Teunissen and Verhagen (2009) and concerns a short kinematic experiment for which the distance between reference and roving receiver varies in the range of 2.5–4.5 km. This will be referred to as a short kinematic baseline. Observations on GPS *L1* and *L2* are available for 3601 epochs. Two different methods of processing are applied, a Kalman filter and an epoch-by-epoch (EBE) processing. In both cases, the ambiguities are fixed every epoch anew.

Table 1 presents the ambiguity resolution probabilities for single- and dual-frequency GPS, indicated with GPS1 and GPS2, respectively. The dual-frequency GPS model is very strong, and hence the ratio test seems to be superfluous for this particular data set since it sometimes results in false alarms while there is no failure, both with Kalman filter and EBE processing. This also applies to the single-frequency Kalman filter results, where the false alarm rates are even higher, whereas without the ratio test ($c = 1$), the failure rate would be zero. Only for the single-frequency EBE results, the ratio test cannot be omitted, since the ILS failure rate is almost 9 %. The failure rate decreases to a value to less than 1 % for $c = 3$. As a consequence, the false alarm rate becomes much higher. It is especially noteworthy that when using the same data, it depends on the processing strategy whether a low or high value of c gives better performance, as can be seen from the single-frequency results obtained with EBE and Kalman processing.

From these results, it can be concluded that the performance of the ratio test depends on the model strength: for strong models, the ratio test might be too conservative inducing unnecessarily high false alarm rates; for weak models, the ratio test may result in too high failure rates depending on the choice of the critical value. In other words: for strong models, a small value for c should be used, while for weak models, a high value is necessary.

Example 2: medium-length baseline

The second example concerns a data set of a 38-km baseline. Data were obtained from the Southern California Integrated GPS Network (<http://www.scign.org>). Single-frequency *L1* and dual-frequency *L1* + *L2* phase and code

observations with 30-s interval are used with a cutoff elevation of 10°. Elevation-dependent weighting is applied with undifferenced standard deviations in zenith direction of 2 mm for phase and 15 cm for code. Ionosphere corrections were determined using four network stations, and a standard deviation, $\sigma_{I,zenith}$, of 1 cm in zenith direction was assigned to these corrections. The zenith troposphere delays are not estimated.

The processing is done either based on Kalman filtering over the whole time span with float ambiguities assumed to be constant and position kinematic, or based on an epoch-by-epoch (EBE) basis. In the latter case, the ambiguities need not to be constant between epochs. In both cases, ambiguity resolution is applied epoch-by-epoch, which means that the ambiguities are fixed every epoch anew.

From Table 2, it follows that with the Kalman filter processing again, the ratio test seems superfluous, since for $c = 1$, all ambiguities are fixed correctly and no false alarms exist. For $c > 1$, however, the ratio test results in many false alarms. On the other hand, if EBE processing is applied, the ratio test does provide protection against wrong fixing, since without the ratio test, wrong ambiguities would be accepted 0.7 % of the time. A critical value of $c = 1.5$ gives the best performance for this particular data set, since with this choice, the false alarm rate is low and the failure rate is reasonably low as well.

Example 3: simulation results dual-frequency GPS

In order to assess the theoretical performance of the ratio test, simulations were used based on the GPS constellation of March 21, 2009. The settings are summarized in Table 3 and Fig. 3. Based on the rule-of-thumb for the relation between $\sigma_{I,zenith}$ and the baseline length given in Odijk (2000), these settings correspond to a baseline of

Table 1 Short baseline in the range of 2.5–4.5 km: Percentage of epochs where a wrong ambiguity is accepted (*failure*) or a *false alarm* is raised by the ratio test

	GPS1		GPS2	
	Failure	False alarm	Failure	False alarm
<i>Kalman</i>				
$c = 1$	0	0	0	0
$c = 1.5$	0	3.9	0	0.7
$c = 2$	0	8.8	0	1.5
$c = 3$	0	12.3	0	3.4
<i>EBE</i>				
$c = 1$	8.7	0	0	0
$c = 1.5$	7.4	19.7	0	0.1
$c = 2$	2.7	34.0	0	0.3
$c = 3$	0.8	48.8	0	2.3

approximately 12.5 km. One million samples of observations were generated using Monte Carlo simulations for each epoch and 300-s interval. Single-epoch ambiguity resolution and two-epoch ambiguity resolution were considered. In the latter case, the ambiguities are assumed time-invariant, but all real-valued parameters are estimated

Table 2 Baseline of 38 km: Percentage of epochs where a wrong ambiguity is accepted (*failure*) or a *false alarm* is raised by the ratio test

	Failure	False alarm
<i>Kalman</i>		
$c = 1$	0	0
$c = 1.5$	0	12.9
$c = 2$	0	20.5
$c = 3$	0	26.9
<i>EBE</i>		
$c = 1$	0.7	0
$c = 1.5$	0.1	1.3
$c = 2$	0.0	5.0
$c = 3$	0	12.5

Table 3 Simulation settings

System	Frequencies	Noise	Troposphere	Locations
GPS	$L1 + L2$	Fig. 2	ZTD estimated	45°N, 75°N

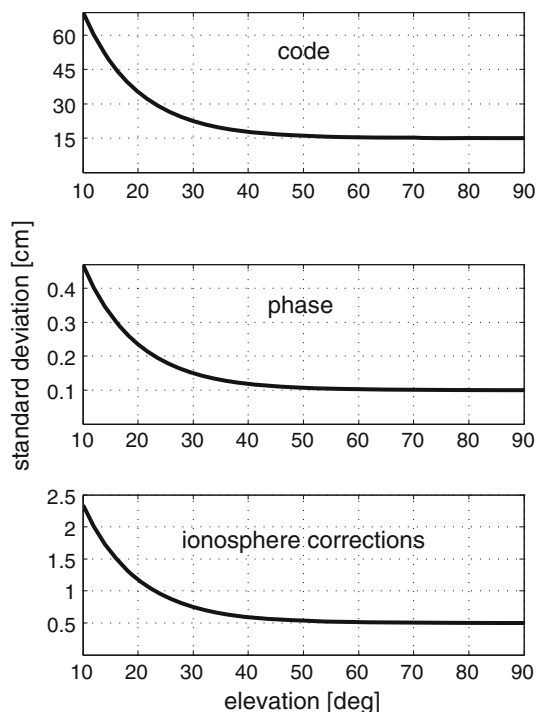


Fig. 2 Standard deviations of undifferenced code and phase observations and the ionosphere corrections as function of satellite elevation

every epoch. The number of satellites and resulting empirical single-epoch failure rates are shown in Fig. 3 for the two locations considered. The ILS failure rate is above 1 % most of the time, with a maximum of 53 %, which is far too high. This shows that testing, with a proper acceptance criterion, is needed so as to get control over the otherwise large failure rate.

Figure 4 presents all failure rates P_f and corresponding false alarm rates P_{fa} obtained during the day at both locations for the different fixed c -values. Inspecting the results for $c = 2$ reveals that the corresponding failure rates vary between 0 and 12 %. With $c = 3$, the failure rates in this specific case will always be well below 0.1 %. In that sense, it could be argued that $c = 3$ is a better choice. On the other hand, it can also be observed that the corresponding false alarm rates will be much higher, reaching 93 %. A very low failure rate in combination with

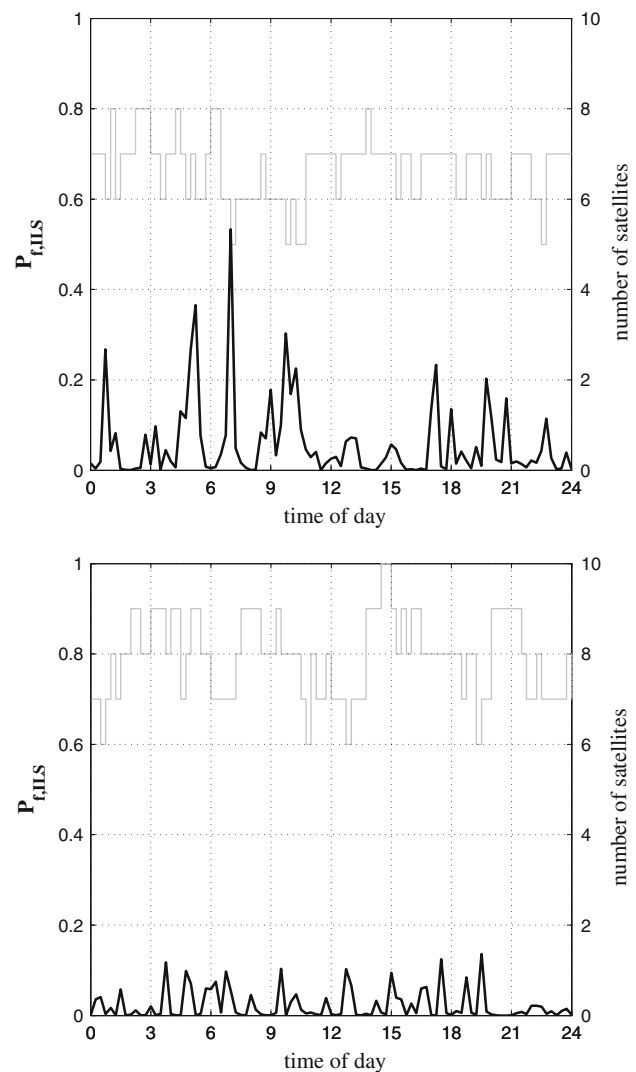


Fig. 3 ILS failure rates and number of satellites with model settings from Table 3 and Fig. 2 for the two locations. Top: 45°N. Bottom: 75°N

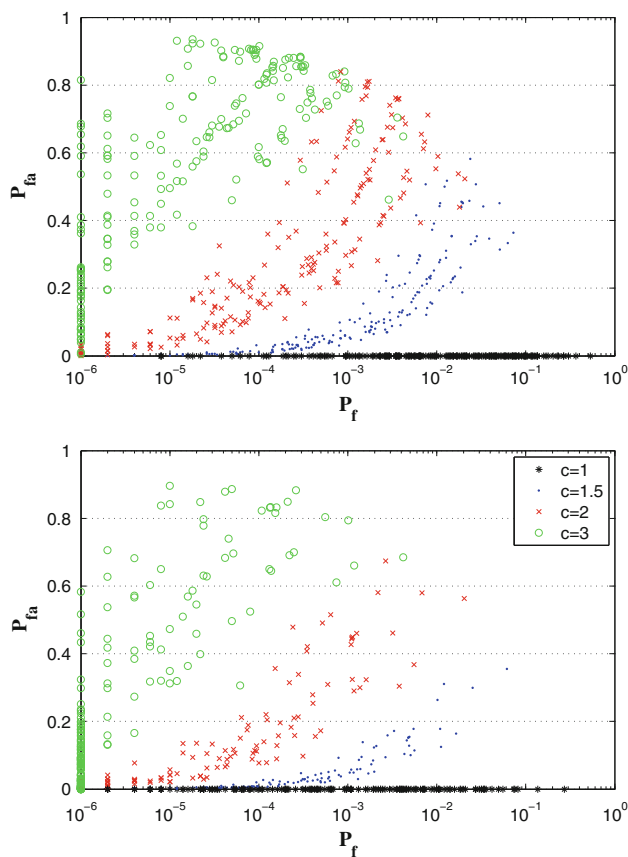


Fig. 4 False alarm rate versus failure rate with different c -values. Each marker corresponds to a model (settings from Table 3 and Fig. 2). *Top*: one epoch. *Bottom*: two epochs

a high false alarm rate is a sign that the critical value is chosen too conservative. However, with a lower value of c , the failure rates may take unacceptably large values at epochs for which the model is weaker due to bad satellite visibility or geometry.

The probabilities obtained with two epochs of data are shown in Fig. 4 as well. Using more epochs will reduce the ILS failure rate, and as a result, both the failure rates and false alarm rates decrease. Still, with $c = 3$, the false alarm rate is generally unnecessarily high, while with $c = 1.5$, there are instances where the failure rate is above 2 %. Comparing the results obtained with one and two epochs, it could be argued that if a fixed c -value was to be used, it should depend on the number of epochs which value would be best to guarantee a low failure rate without being too conservative.

Performance of ratio test with fixed critical value

The three examples presented here show that the performance of the ratio test with a fixed c -value depends on the processing strategy, that is, Kalman filter or EBE, and on

the model strength of the given application. The model strength depends on receiver satellite geometry, number of frequencies and epochs, measurement noise, baseline length, and the availability and accuracy of the troposphere and ionosphere corrections.

If we make a general distinction between ‘strong’ and ‘weak’ models, it follows that for strong models, the fixed c -values currently in use are often too conservative, so that the false alarm rates are unnecessarily high, while the failure rates are very close to zero. For weak models, the fixed c -values are often too low, so that the fixed solution is often wrongly accepted, thus resulting in high failure rates.

These conclusions suggest that a model-driven c -value should be used such that the failure rate can be controlled (Teunissen 2003, 2005) and at the same time ensuring that the test is not overly conservative.

The fixed-failure rate ratio test

The problem of using the ratio test with a fixed c -value is that depending on model strength, either the failure rate or the false alarm rate can become too high. Since incorrect integer estimation may have a severe impact on the baseline solution, it is especially desirable to have control over the failure rate. Therefore, the ratio test with fixed failure rate was introduced (Verhagen and Teunissen 2006; Teunissen and Verhagen 2009). For this test, a model-driven c -value is used such that it guarantees that the failure rate does not exceed a user-defined threshold. This test is called the fixed-failure rate ratio test (FF-RT). Tandy and Young (2012) proposed a modified version of the fixed failure rate approach, which includes an indication of how many epochs are required.

The advantage of our FF-RT is that the user can now control the failure rate. Similar to relation (3), it can be seen from Fig. 1 that the following relations hold,

$$P_f \nabla \Rightarrow c \blacktriangle; P_{fix} \nabla; P_{fa} \blacktriangle \quad (4)$$

Hence, a smaller choice for the fixed failure rate implies that c will become larger and the test becomes more restrictive, leading to lower fixing rates and higher false alarm rates. The performance of the fixed-failure rate ratio test will now be analyzed for the same examples as before and compared with the fixed c -value results.

Example 1: short kinematic baseline

Table 4 shows the short kinematic baseline FF-RT results for a failure rate fixed at 0.1 %. As compared to the results obtained with the fixed c -values, it can be observed that for the strong models, the performance of the ratio test is better

Table 4 FF-RT results with $P_f = 0.1\%$ for a short baseline of 2.5–4.5 km in length

	GPS1		GPS2	
	Failure	False alarm	Failure	False alarm
Kalman	0	0.0	0	0
EBE	0.1	58.8	0	0.0

Percentage of epochs where a wrong ambiguity is accepted (*failure*) or a *false alarm* is raised by the ratio test (compare with Table 1)

Table 5 FF-RT results with $P_f = 0.1\%$ for a 38-km baseline

	Failure	False alarm
Kalman	0	0.9
EBE	0.1	2.9

Percentage of epochs where a wrong ambiguity is accepted (*failure*) or a *false alarm* is raised by the ratio test. (compare with Table 2)

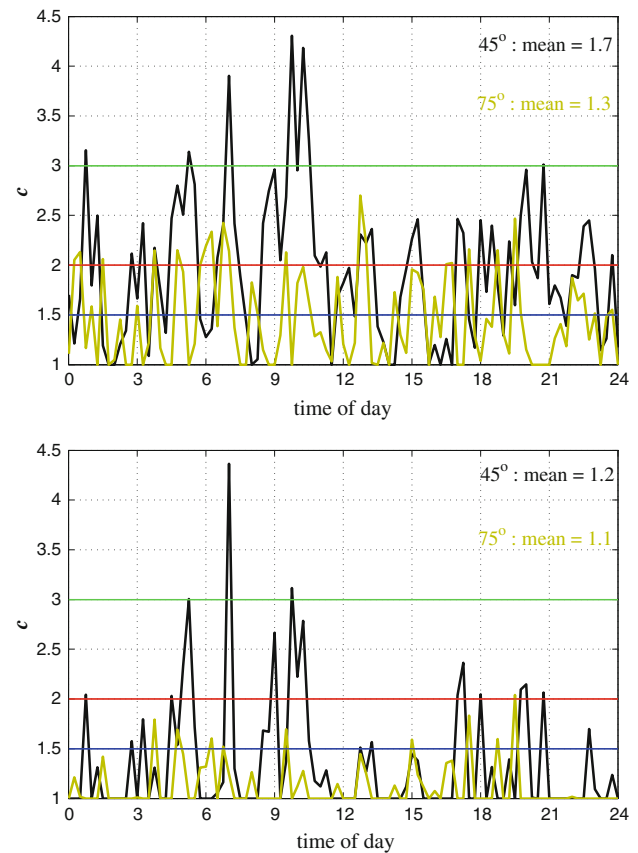
in the sense that it does not lead to unnecessary false alarms. For the weak model, that is, based on single-frequency EBE processing, the false alarm rate is now higher than with the fixed c -values. However, this is acceptable, since at the same time, the failure rate is indeed below the target value of 0.1 %.

Example 2: medium-length baseline

Table 5 shows the results for the medium-length baseline of 38 km, which should be compared with Table 2. For the EBE processing, a fixed value of $c = 1.5$ gives comparable results as the FF-RT approach: the failure rate is only slightly larger than 0.1 %, whereas the number of false alarms is somewhat lower. However, with the Kalman filter processing, all three choices for the threshold c are overly conservative; even with $c = 1.5$, a false alarm is raised for almost 13 % of the time, while with the fixed failure rate, it is less than 1 % of the time.

Example 3: simulation results dual-frequency GPS

Figure 5 presents the c -values obtained with the fixed failure rate approach for the simulation results. See Table 3; Figs. 2 and 3. The fixed c -values as currently used in practice are also shown. Single-epoch results are in the top panel for both locations, and results with two epochs are in the bottom panel. Large fluctuations in c -value can be observed depending on satellite visibility and geometry; the fluctuations follow the same pattern as the number of visible satellites and/or ILS failure rates as shown in Fig. 3. The number of epochs clearly has an impact as well: with

**Fig. 5** c -values with fixed failure rate of 0.1 % (cases from Fig. 2 and Table 3). *Top*: one epoch. *Bottom*: two epochs

two epochs, the c -values are smaller, but still values above 3 are obtained occasionally.

At 75°N on average more satellites are visible, so that the ILS failure rates are generally smaller (see Fig. 3). As a consequence, the c -values obtained with the fixed failure rate approach are on average smaller compared to those obtained at 45°N.

Note that the *mean* c -values are below 2 in all cases. However, if higher measurement noise or more uncertainty for the ionosphere corrections would have been considered, the corresponding c -values would have been higher.

Figure 6 shows the false alarm rates P_{fa} and fix probabilities P_{fix} as obtained with the fixed c -values minus the corresponding probabilities obtained with the fixed failure rate approach. Only the single-epoch results are shown, since the results for the two-epoch case look very similar.

It can be observed that if the failure rate with the fixed c -value happens to be equal to 0.1 %, the two approaches will have the same outcome and the differences in false alarm and fix probability will be zero, as is visible in Fig. 6. Furthermore, if the failure rate obtained with using a fixed c -value is below 0.1 %, the corresponding false alarm rate is always higher than with the FF approach. This

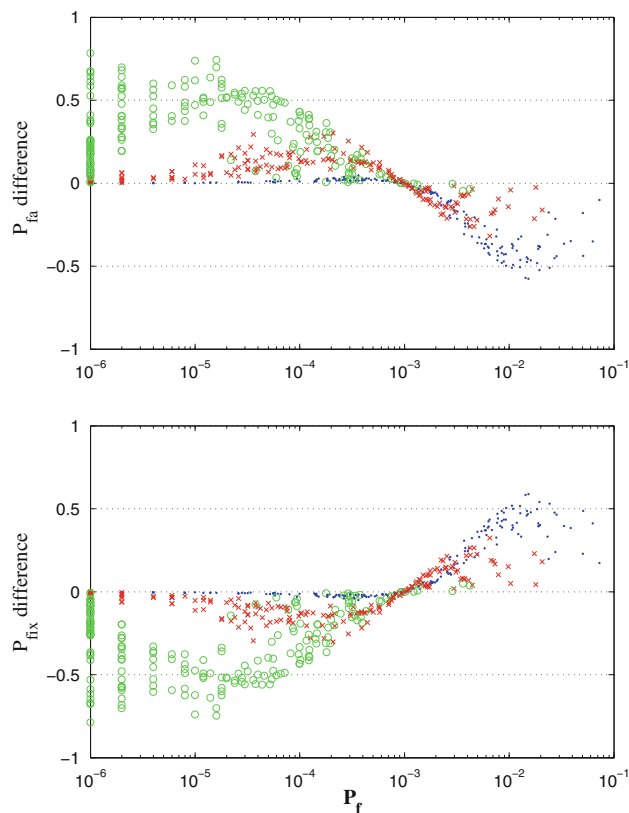


Fig. 6 Probability differences as function of the failure rate obtained with fixed c -value (shown in Fig. 4); only single-epoch results are shown. Top: $P_{faFC} - P_{faFF}$. Bottom: $P_{fixFC} - P_{fixFF}$ (FC = fixed c ; FF fixed failure rate)

indicates that the fixed c -values are in fact too conservative and the fix probability with the FF approach will be higher.

Conversely, if the failure rate with fixed c -value is above 0.1 %, which is undesirable, the false alarm rate with the FF approach will be higher, and the fix probability will be lower. This is the compromise that has to be made in order to prevent that the failure rate gets too high.

Finally, it can be seen that with increasing fixed c -value, the failure rate will be decreasing, and at the same time, the false alarm rate increasing (relation (3), see also Fig. 4), which explains why the difference in false alarm rate is larger with increasing c -value, and smaller for the fix rate.

Relation between c and GNSS model parameters

A practical disadvantage of the fixed failure rate approach is that no analytical expression exists that can be used to directly compute the appropriate c -value for a given failure rate. In the “Appendix”, a simulation-based approach is described based on Verhagen (2005a), which is, however, computationally expensive. It is therefore interesting to investigate whether a relation can be found between the

model parameters and the model-driven c -value such that the failure rate is below the required threshold.

In this section, it will be shown that such a relation is indeed present, albeit that it is not a one-to-one relation that allows ‘exact’ computation of the c -value for a given set of GNSS model parameters.

First, the results using the simulation-based approach with 100,000 samples will be presented based on different models using each combination of the settings in Table 6. If GPS and Galileo combination is considered, one reference satellite per system is selected in order to take into account the Inter-System Biases (Odijk et al. 2012). The values of $\sigma_{I,zenith}$ correspond to baseline lengths of approximately 12.5 and 60 km, respectively (Odijk 2000).

Figure 7 presents the mean, minimum and maximum c -value, as well as 20 and 80 percentile as function of the number of satellites. A fixed failure rate of 0.1 % was applied. Results obtained with the different systems and locations are combined, using the settings from Table 6. Note that in each of the panels, the model strength varies solely due to the different satellite geometries. The results can be interpreted as follows:

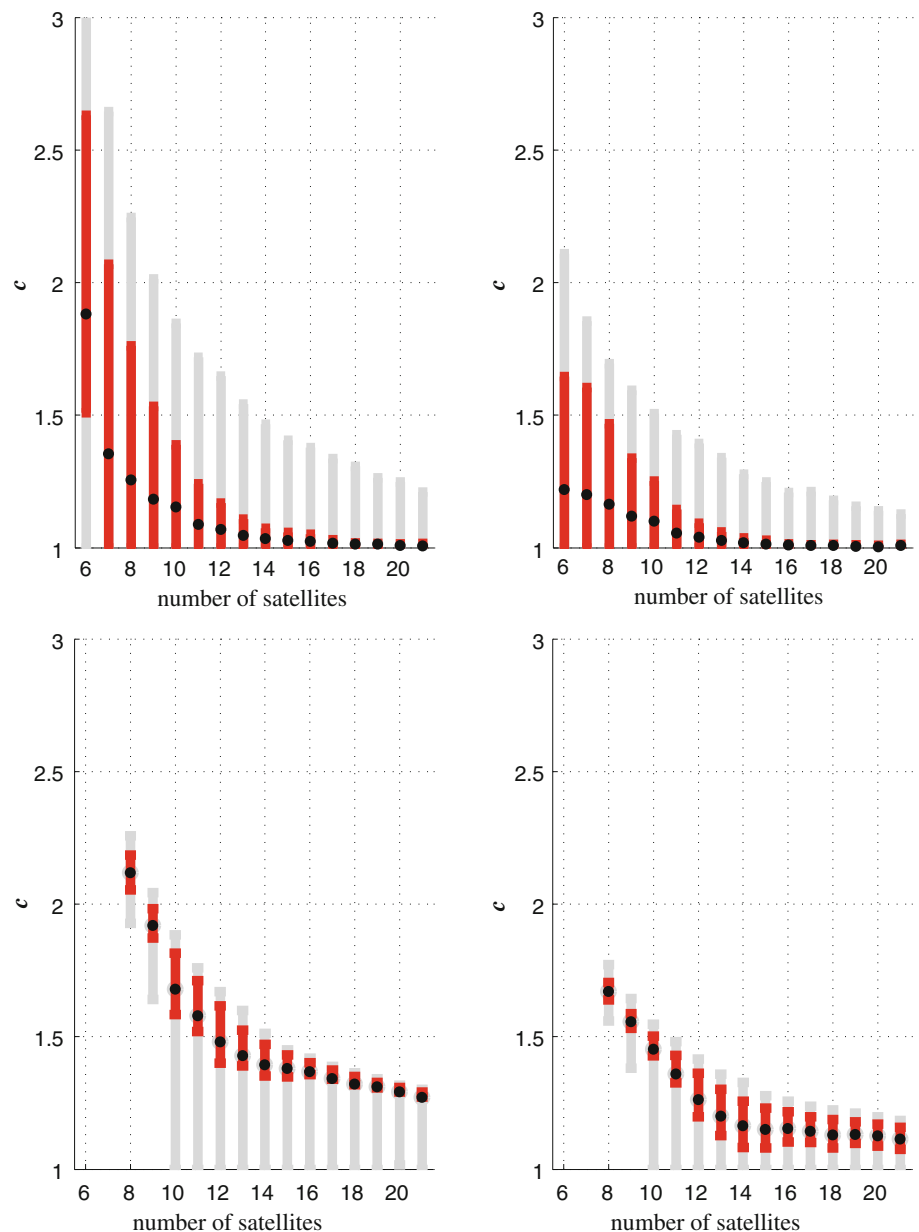
- For example, with two frequencies and $\sigma_{I,zenith} = 5$ mm (top left panel), the mean c -value for six visible satellites is approximately equal to 1.9. However, during 20 % of the time, the appropriate c -value is larger than 2.7 with a maximum of 4, and in 20 % of the time, the value is smaller than 1.5 with a minimum of 1.
- The relations between the c -value and the number of satellites are that the c -value decreases with increasing number of satellites, and in all cases, it converges to a certain value for larger numbers of satellites. For shorter baselines (smaller $\sigma_{I,zenith}$), c can be chosen equal to 1 during more than 80 % of the time if the number of visible satellites is high, as will be the case with combined GPS and Galileo. The ratio test can then be discarded.

Table 6 Simulation settings (results in Fig. 7)

System	GPS (nominal constellation) Galileo GPS + Galileo
Times	1,250 different epochs in 10-day period
Frequencies	$L1 + L5$ $L1 + L5 + L2/E5b$
Phase and code noise	See Fig. 2
Atmosphere conditions	$\sigma_{I,zenith} = 5$ or 25 mm; ZTD estimated
Locations	60°S, 30°S, 0°N, 40°N, 80°N,

$\sigma_{I,zenith}$ is the standard deviation of the zenith ionosphere corrections

Fig. 7 Single-epoch ratio test statistics for critical value c , with $P_f = 0.1\%$. The mean c -value is shown with a black marker. The gray bars show the complete range of values c may take as function of the number of visible satellites. The red bars are delimited by the 20 and 80 percentiles. *Left column*: two-frequency case; *Right column*: three-frequency case. *Top*: $\sigma_{I,zenith} = 5$ mm; *Bottom*: $\sigma_{I,zenith} = 25$ mm



- c. More frequencies will result in a stronger model, such that the c -value will be smaller. The impact of an additional frequency is especially significant for lower number of satellites.
- d. More uncertainty in the ionosphere corrections, that is, a larger $\sigma_{I,zenith}$, will result in a weaker model. Consequently, the c -values will become larger for longer baselines.

Interestingly, the gray and red bars are shorter in the bottom panels, which means that the c -value will generally be in a smaller range for a given number of satellites. Note that in the bottom panels, no results are shown for six and seven visible satellites. In those cases, the ILS failure rate was nearly always $\gg 25\%$ and the FF-RT fix rate became

very small ($\sim 1\%$). Therefore, no fixing attempt was made.

Careful inspection of the results in Fig. 7 reveals that the c -value from the FF-RT will be smaller than the fixed values of 1.5, 2 and 3 if many satellites are visible. More specifically, if more than 12 satellites are visible, the fixed c -values are too conservative for these particular models, since the model-based c -values with FF-RT are nearly always smaller. On the other hand, for smaller numbers of satellites, there can be a large probability that the fixed c -values of 1.5 and 2 are in fact too small. For example, for the dual-frequency scenario in the top panel with six visible satellites, there is a 63 % chance that a value of $c = 2$ is too small, leading to a high failure rate.

It should be mentioned that if the c -value would have been plotted as a function of the number of ambiguities rather than number of satellites, the panels on the left and right could be combined and then show that in fact a clear relation between c -value and number of ambiguities is present. Hence, the results show that there is a relation between the c -value and the number of satellites and frequencies and thus the number of unknown ambiguities. These results, however, have been obtained for only a few settings of the noise level (Table 6). To include variations in these noise levels as well, we now also take the ILS failure rate into account when considering the relation between the c -values and the number of ambiguities n . The ILS failure rate is here chosen as parameter since it captures the model strength in its dependence on all contributing factors.

The relations between the c -value, the number of ambiguities n and the ILS failure rate $P_{f,ILS}$ are shown in Fig. 8 for all the cases given in Table 7. Compared to the case in Table 6, more choices for the measurement noise and atmosphere conditions are considered. Each color in Fig. 8 represents a number of ambiguities n . The corresponding c -values are plotted as function of the ILS failure rate for every model. Note, since the nominal GPS constellation has fewer satellites than Galileo, that the average c -value is smaller for Galileo, that is, independent from n .

Creation and usage of look-up tables

In the “Appendix,” it is explained how results like those of Fig. 8 can be obtained and used to create look-up tables for a given fixed failure rate. Such a table is available in the new version 3.0 of the freely available LAMBDA software for fixed failure rates of 0.1 and 1 %, respectively.

The concept is illustrated in Fig. 9 for the case $n = 24$: each dot represents the appropriate c -value as function of the ILS failure rate for all cases where $n = 24$. The black line is the ‘conservative’ relation between c and $P_{f,ILS}$. It is conservative in the sense that the simulation-based c (shown by the dots) is always lower than or equal to the value obtained from the relation (shown by the line), hence resulting in a somewhat lower failure rate and fix rate.

For weak models, the value for c will be large, and as a consequence, the fix probability will be very low. This becomes clear from Fig. 10, where the fix probabilities as function of ILS failure rate are shown corresponding to the results in Fig. 8. It follows that the fix probability will generally be less than 10 % if the ILS failure rate is above 20 %. Therefore, a user can choose not to make a fixing attempt if the ILS failure rate is above a certain value.

Table 8 shows an example of a look-up table for the fixed failure rate of 0.1 %, which was created using the

approach described in the “Appendix”. Note that here the reciprocal ratio test is applied: the threshold values in the table are $\mu = 1/c$ as function of the ILS failure rate and the dimension n . In this table, we have that the maximum ILS failure rate is set to $P_{f,ILS} = 0.25$ (i.e., 25 %).

The FF-RT approach guarantees that the failure rate will be below a given threshold. It does therefore not make sense to choose a maximum for the threshold value $\mu = 1/c$ such that the fixed solution is rejected if the float solution is close to the boundary of the pull-in region. If one finds that the failure rate is higher than expected, this is an indication that something is wrong with the model. For example, if the measurement noise is chosen too optimistic, this will result in a too large acceptance region.

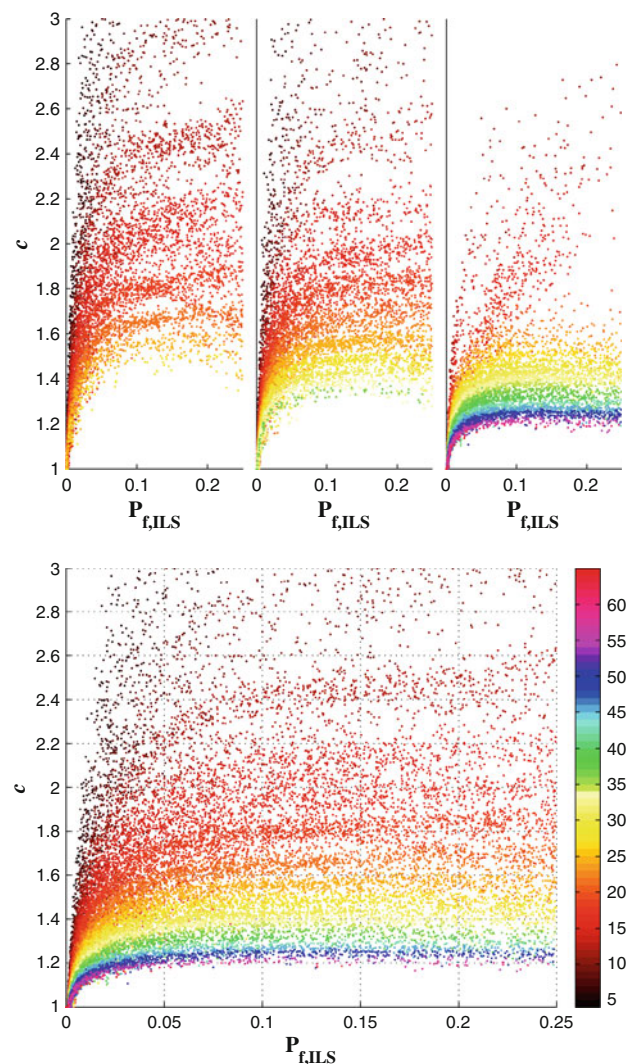
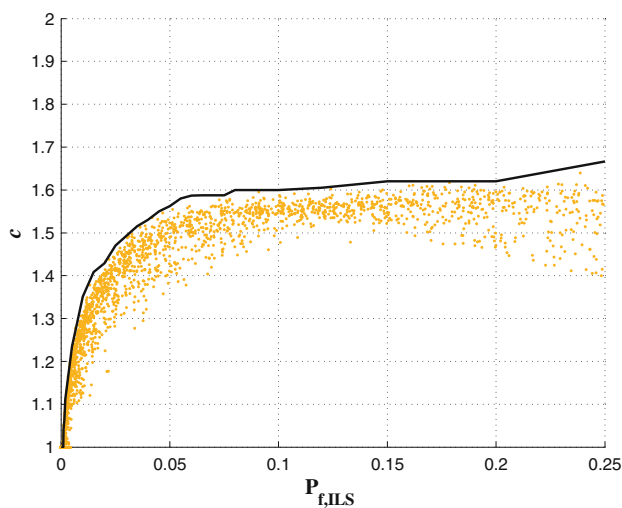
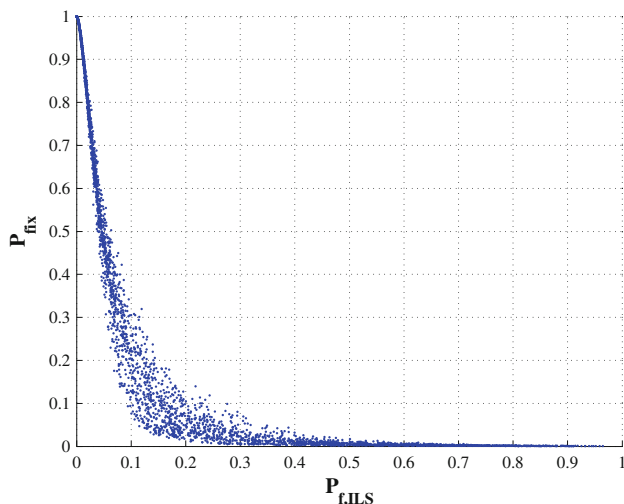


Fig. 8 Relation between number of ambiguities (color scale), ILS failure rate and c -value for $P_f = 0.1$ %. Top panels show from left to right the results for GPS, Galileo and GPS + Galileo, respectively. The bottom panel shows the results for all three systems

Table 7 Simulation settings

System	GPS (nominal constellation) Galileo GPS + Galileo
Times	101 different epochs in 10-day period
Frequencies	$L1 + L5$ $L1 + L5 + L2/E5b$
Phase and code noise	See Fig. 2 with scale factors of 0.25, 0.5, 1, 2, 4
Atmosphere conditions	$\sigma_{I,zenith} = 0$ mm; ZTD not estimated $\sigma_{I,zenith} = 5, 10, 25, 50$ mm; ZTD estimated
Locations	60°S, 30°S, 0°N, 40°N, 80°N,

**Fig. 9** Relation between ILS failure rate and c -value for $n = 24$. Simulation results (dots) and approximate values determined with look-up table (line) are shown**Fig. 10** Relation between ILS failure rate and fix probability

Creation of your own look-up table

The above results have been produced using many different models and through the variation of all those factors that have an impact on model strength. Certain users may, however, work with a specific receiver at a certain location, such that, for instance, the GNSS system, the number of frequencies and measurement noise are always the same. In that case, tailor-made look-up tables and corresponding graphs can be created.

The procedure for creating tailor-made look-up tables is similar to that as described above. The variation in the simulations, however, need now only consider those aspects that are relevant to the specific user application. Thus, step 1 of Appendix 2 is now only executed for those models relevant to the application. For this model simulation, the VISUAL software can be used (Verhagen 2002). Yuma almanacs (<http://navcen.uscg.gov/?pageName=gpsAlmanacs>) or broadcast navigation data may be used to calculate the satellite positions at different epochs. Together with the values for the measurement noise and the assumptions on the atmosphere parameters, the matrices A and B of model (1), as well as Q_{yy} for each epoch, can be constructed. They are used to determine the variance matrix of the float ambiguities, $Q_{\hat{a}\hat{a}}$. For each such variance matrix, the simulation is executed. The remaining procedure is then identical as described in the “Appendix,” thus resulting in a look-up table tailored to the cases in which a user is interested.

The ratio test is now also included as an option in the Matlab implementation version 3.0 of the LAMBDA software. It is available from the authors upon request (<http://saegnss2.curtin.edu.au/~gnssweb/index.php?request=getlambda>).

Concluding remarks

The popular ratio test for deciding whether or not to fix the carrier phase ambiguities was reviewed. Our results show that the traditional use of the ratio test, that is, with a fixed threshold c -value often results in either unacceptably high failure rates or is overly conservative. It follows that for strong models, the fixed c -values currently in use are often too conservative, so that the false alarm rates are unnecessarily high, while the failure rates are very close to zero. For weak models, on the other hand, the currently used fixed c -values are often too low, so that the fixed solution is often wrongly accepted, thus resulting in high failure rates.

These problems can be overcome if the ratio test is made adaptive to the strength of the underlying GNSS model. We therefore proposed to replace the current fixed threshold approach by the more flexible fixed failure rate approach.

Table 8 Portion of look-up table for $P_f = 0.001$ (0.1 %) with the threshold values $\mu = 1/c$

$P_{f,ILS}$	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	...
0.0000	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	...
0.0010	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	...
0.0012	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.94	0.96	0.96	0.96	0.96	0.96	0.96	0.96	...
0.0015	0.87	0.87	0.88	0.88	0.89	0.89	0.89	0.89	0.91	0.92	0.92	0.92	0.92	0.93	0.93	...
0.0020	0.78	0.78	0.80	0.80	0.81	0.82	0.83	0.84	0.84	0.86	0.86	0.86	0.87	0.88	0.88	...
0.0050	0.54	0.54	0.57	0.57	0.59	0.64	0.64	0.68	0.68	0.69	0.71	0.72	0.73	0.74	0.75	...
0.0100	0.36	0.38	0.41	0.43	0.45	0.51	0.52	0.57	0.57	0.59	0.61	0.62	0.64	0.65	0.67	...
0.0150	0.27	0.29	0.32	0.36	0.38	0.43	0.46	0.51	0.52	0.53	0.55	0.56	0.58	0.60	0.62	...
0.0200	0.21	0.24	0.27	0.30	0.33	0.38	0.42	0.46	0.48	0.49	0.52	0.52	0.55	0.57	0.59	...
0.0250	0.19	0.22	0.24	0.27	0.29	0.35	0.39	0.42	0.45	0.46	0.49	0.49	0.53	0.54	0.56	...
0.0300	0.17	0.20	0.22	0.24	0.26	0.32	0.35	0.40	0.42	0.44	0.47	0.47	0.51	0.53	0.54	...
0.0350	0.16	0.18	0.20	0.22	0.24	0.30	0.33	0.38	0.40	0.42	0.45	0.45	0.49	0.51	0.53	...
0.0400	0.15	0.17	0.18	0.21	0.22	0.27	0.31	0.36	0.39	0.40	0.44	0.44	0.48	0.50	0.52	...
0.0450	0.14	0.16	0.17	0.19	0.21	0.26	0.29	0.34	0.38	0.39	0.43	0.44	0.48	0.49	0.51	...
0.0500	0.13	0.15	0.16	0.18	0.19	0.24	0.28	0.32	0.37	0.39	0.42	0.43	0.47	0.48	0.51	...
0.0550	0.12	0.14	0.15	0.17	0.18	0.23	0.27	0.31	0.36	0.38	0.41	0.42	0.46	0.48	0.50	...
0.0600	0.12	0.13	0.15	0.16	0.17	0.21	0.26	0.30	0.36	0.37	0.40	0.42	0.46	0.47	0.50	...
0.0650	0.11	0.12	0.14	0.15	0.16	0.20	0.26	0.30	0.35	0.36	0.40	0.41	0.45	0.46	0.50	...
0.0700	0.11	0.12	0.13	0.14	0.15	0.20	0.25	0.30	0.35	0.36	0.39	0.41	0.45	0.46	0.49	...
0.0750	0.10	0.11	0.12	0.13	0.15	0.19	0.24	0.29	0.34	0.36	0.39	0.41	0.44	0.45	0.49	...
0.0800	0.10	0.10	0.12	0.13	0.14	0.19	0.23	0.29	0.34	0.35	0.38	0.40	0.44	0.45	0.49	...
0.0850	0.10	0.10	0.11	0.12	0.14	0.18	0.23	0.28	0.33	0.35	0.38	0.40	0.44	0.45	0.49	...
0.0900	0.09	0.09	0.11	0.12	0.13	0.18	0.22	0.28	0.33	0.35	0.38	0.40	0.43	0.45	0.49	...
0.0950	0.09	0.09	0.10	0.11	0.13	0.18	0.22	0.28	0.33	0.34	0.38	0.40	0.43	0.45	0.48	...
0.1000	0.09	0.09	0.10	0.11	0.12	0.17	0.22	0.27	0.32	0.34	0.37	0.40	0.43	0.44	0.48	...
0.1200	0.08	0.08	0.09	0.09	0.11	0.16	0.21	0.26	0.31	0.33	0.37	0.39	0.42	0.44	0.48	...
0.1500	0.07	0.07	0.08	0.08	0.10	0.16	0.20	0.25	0.29	0.32	0.35	0.39	0.41	0.44	0.47	...
0.2000	0.05	0.05	0.06	0.06	0.10	0.15	0.19	0.24	0.27	0.31	0.34	0.38	0.41	0.43	0.46	...
0.2500	0.04	0.04	0.05	0.06	0.09	0.14	0.18	0.23	0.25	0.30	0.34	0.37	0.41	0.43	0.46	...
0.2600	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	...
1.0000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	...

The number in the header (columns 2–16) equals n

With this approach, the user is given control over the failure rate for his particular application.

Our simulation results have shown that it is possible to establish empirical relations between the appropriate c -value and GNSS model parameters, such as receiver satellite geometry, number of frequencies, number of measurement epochs, measurement noise, baseline length and atmospheric circumstances.

From these results follows, for instance, that with more observations due to the availability of more frequencies and/or GNSSs, the c -values can be chosen much smaller than the currently used fixed values. Furthermore, it is interesting to note that for larger n , the appropriate c -value is stabilizing with increasing ILS failure rate (Fig. 8) and for a given n can be chosen constant for ILS failure rates

above say 5 %. For very strong models, the value c becomes even equal to 1 when $P_f = 0.1$ %. In that case, one can do without the ratio test or choose a more stringent, and thus smaller, fixed failure rate.

To perform the simulations, we also provided the numerical procedure for determining the fixed failure rate c -values. Additionally, we described the procedure for creating general or tailor-made look-up tables for these fixed failure rate c -values.

As mentioned above, the ratio test is only one member from the class of integer aperture estimators. Hence, the method of evaluation as presented can be applied to these other tests as well. Such work would then also be able to compare the performance of these tests and answer the question which of the traditional tests such as ratio test,

difference test or projector test performs best. It would then also become possible to compare their performance with the optimal test, that is, the one that maximizes the success rate for a given fixed failure rate (Teunissen 2004, 2005). Since no widespread experience yet exists with this optimal test, an approach similar to the one presented can be used to investigate its characteristics.

Acknowledgments Peter Teunissen is Federation Fellow of the Australian Research Council (project FF0883188). The research of Sandra Verhagen is supported by the Dutch Technology Foundation STW, applied science division of NWO and the Technology Program of the Ministry of Economic Affairs. Part of this work was done in the framework of the project ‘New Carrier Phase Processing Strategies for Next Generation GNSS Positioning’ of the Cooperative Research Centre for Spatial Information (CRC-SI2). All this support is gratefully acknowledged.

Appendix

Here, we will work with the reciprocal of the ratio test, which means that $\tilde{a}$ is accepted if $\frac{q(\tilde{a})}{q(\hat{a})} \leq \frac{1}{c} = \mu$, since the threshold value μ is then limited as $0 \leq \mu \leq 1$.

Procedure for determining the threshold value μ based on simulations

1. For a given model, calculate $\mathbf{P}_{f,ILS}$, for example, use the upper bound from integer bootstrapping (Teunissen 1999). If $\mathbf{P}_{f,ILS} \leq \mathbf{P}_f$, set $\mu = 1$, otherwise continue with step 2.
2. Using Monte Carlo simulations, generate N samples of normally distributed float ambiguities: $\hat{\mathbf{a}}_i \sim N(0, \mathbf{Q}_{\hat{\mathbf{a}}\hat{\mathbf{a}}})$, $i = 1, \dots, N$.
3. Determine the ILS solutions (best and second-best candidates) $\tilde{\mathbf{a}}_i, \tilde{\mathbf{a}}'_i$, $R_i = \frac{q(\tilde{\mathbf{a}}_i)}{q(\tilde{\mathbf{a}}'_i)}$.
4. The simulation-based failure rate as function of μ is given by $\mathbf{P}_f(\mu) = \frac{N_f}{N}$ with $N_f = \sum_{i=1}^N \omega(R_i, \tilde{\mathbf{a}}_i)$ where $\omega(R_i, \tilde{\mathbf{a}}_i)$ equals 1 if $R_i \leq \mu$, $\tilde{\mathbf{a}}_i \neq 0$, and 0 otherwise.

Choose $\mu_{\min} = (\min(R_i) - 10^{-16})$ since this results in $N_f = 0 \Rightarrow \mathbf{P}_f(\mu_{\min}) = 0$, $\mu_{\max} = \max(R_i)$ which results in $\mathbf{P}_f(\mu_{\max}) = \mathbf{P}_{f,ILS}$.

5. Use a root finding method to find $\mu \in [\mu_{\min}, \mu_{\max}]$ so that $\mathbf{P}_f(\mu) - \mathbf{P}_f = 0$.

This will give the solution since $\mathbf{P}_f(\mu_{\min}) - \mathbf{P}_f < 0$ and $\mathbf{P}_f(\mu_{\max}) - \mathbf{P}_f > 0$,

and the failure rate is monotonically increasing for increasing μ .

Procedure for creating the look-up table

1. Generate many different models based on varying satellite geometry (system, time and location), number of frequencies, number of epochs, measurement noise, baseline length and the availability and accuracy of atmosphere corrections.
2. For each model, use the simulation approach described above to find the threshold value for the given fixed failure rate $\mathbf{P}_f$.
3. Select all models with the same number of ambiguities n .
4. Plot the ILS failure rate $\mathbf{P}_{f,ILS}$ versus the threshold value μ determined with the simulations for all models with the same n (See example in Fig. 9. Note that here $c = 1/\mu$ is shown).
5. In order to be conservative, one can manually select the points creating the blue solid line as in Fig. 9, which ensures that with the corresponding threshold value for a given ILS failure rate, this value $\mu = 1/c$ is somewhat lower than required, that is, more conservative.
6. The solid blue line can then be used to determine the appropriate threshold values μ for the ILS failure rates in the table, and these can be stored in the column corresponding to dimension n .

References

- Abidin HA (1993) Computational and geometrical aspects of on-the-fly ambiguity resolution. Dissertation, Department of Surveying Engineering, Technical Report No 104, University of New Brunswick, Canada
- Chen Y (1997) An approach to validate the resolved ambiguities in GPS rapid positioning. In: Proceedings of the International symposium on kinematic systems in geodesy, geomatics and navigation, Banff, Canada, pp 301–304
- Euler HJ, Schaffrin B (1990) On a measure for the discernibility between different ambiguity solutions in the static-kinematic GPS-mode. In: IAG Symposia no.107, kinematic systems in geodesy, surveying, and remote sensing, Springer-Verlag, New York, pp 285–295
- Frei E, Beutler G (1990) Rapid static positioning based on the fast ambiguity resolution approach FARA: theory and first results. Manuscripta Geodaetica 15(6):325–356
- Ha S (1997) Quality control issues relating to instantaneous ambiguity resolution for real-time GPS kinematic positioning. J Geodesy 71(6):351–361
- Han S, Rizos C (1996a) Validation and rejection criteria for integer least-squares estimation. Survey Review 33(260):375–382
- Han S, Rizos C (1996b) Integrated methods for instantaneous ambiguity resolution using new-generation GPS receivers. In: Proceedings of the IEEE PLANS'96, Atlanta GA, pp 245–261
- Landau H, Euler HJ (1992) On-the-fly ambiguity resolution for precise differential positioning. In: Proceedings of the ION GPS-1992, Albuquerque NM, pp 607–613

- Odijk D (2000) Weighting ionospheric corrections to improve fast GPS positioning over medium distances. In: Proceedings of the ION GPS-2000, Salt Lake City UT, pp 1113–1123
- Odijk D, Teunissen PJG, Huisman L (2012) First results of mixed GPS + GIOVE single-frequency RTK in Australia. *J Sp Sci* 57:3–18
- Parkins A (2011) Increasing GNSS RTK availability with a new single-epoch batch partial ambiguity resolution algorithm. *GPS Solut* 15:391–402
- Tandy M, Young KW (2012) Variable duration fixed failure rate ambiguity resolution. *GPS Solut*. doi:10.1007/s10291-012-0265-9
- Teunissen PJG (1995) The least-squares ambiguity decorrelation adjustment: a method for fast GPS ambiguity resolution. *J Geodesy* 70(1–2):65–82
- Teunissen PJG (1999) An optimality property of the integer least-squares estimator. *J Geodesy* 73(11):587–593
- Teunissen PJG (2003) Integer aperture GNSS ambiguity resolution. *Artif Satell* 38(3):79–88
- Teunissen PJG (2004) GNSS ambiguity resolution. *J Geodesy* 78:235–244
- Teunissen PJG (2005) Penalized GNSS ambiguity resolution with optimally controlled failure-rate. *Artif Satell* 40(4):219–227
- Teunissen PJG, Verhagen S (2009) The GNSS ambiguity ratio-test revisited: a better way of using it. *Survey Rev* 41(312):138–151
- Tiberius CCJM, De Jonge PJ (1995) Fast positioning using the LAMBDA method. In: Proceedings of the DSNS'95, Bergen, Norway
- Verhagen S (2002) Studying the performance of global navigation satellite systems: a new software tool. *GPS World* 13(6):60–65
- Verhagen S (2004) Integer ambiguity validation: an open problem? *GPS Solut* 8(1):36–43
- Verhagen S (2005a) The GNSS integer ambiguities: estimation and validation. Dissertation, Publications on Geodesy, 58, Netherlands Geodetic Commission, Delft
- Verhagen S (2005b) On the reliability of integer ambiguity resolution. *Navigation* 52(2):99–110
- Verhagen S, Teunissen PJG (2006) New global navigation satellite system ambiguity resolution method compared to existing approaches. *J Guid Cont Dyn* 29(4):981–991
- Wang J, Stewart MP, Tsakiri M (1998) A discrimination test procedure for ambiguity resolution on-the-fly. *J Geodesy* 72(11):644–653
- Wei M, Schwarz KP (1995) Fast ambiguity resolution using an integer nonlinear programming method. In: Proceedings of the ION GPS-1995, Palm Springs CA, pp 1101–1110

Author Biographies

Sandra Verhagen obtained her PhD from and is now an assistant professor at Delft University of Technology. Her research focuses on ambiguity resolution and integrity monitoring for real-time kinematic GNSS applications.

Peter J. G. Teunissen is a Federation Fellow of the Australian Research Council (ARC), Professor of Geodesy and Navigation and Head of CUT's Global Navigation Satellite System (GNSS) Research Lab. His current research focus is on modelling next-generation GNSS for relative navigation and attitude determination in space and air.